Intelligent Agents and Search Strategies

Artificial Intelligence - Unit 1-2 - Problem solving

What is AI?

Artificial intelligence studies how to build systems that perceive their environment and take actions that maximize their chance of achieving goals. Typical areas include search, knowledge representation, machine learning, natural language processing and computer vision.

Intelligent agents

- An agent maps percepts from sensors to actions on actuators.
- Types: simple reflex, model-based reflex, goal-based, utility-based and learning agents.
- Environments are described as observable or partial, deterministic or stochastic, static or dynamic, discrete or continuous.

State-space search

Problems are formulated with an initial state, actions, a transition model, a goal test and a path cost.

- BFS: complete, optimal for equal step costs, O(b^d) memory.
- DFS: low memory O(bm) but not complete on infinite or cyclic spaces.
- Uniform-cost search: expands the cheapest path first; optimal.
- Iterative deepening: DFS memory with BFS completeness.

Informed search and A*

A heuristic h(n) estimates the cost from node n to the goal. A* expands the node with the lowest f(n) = g(n) + h(n). It is optimal when h is admissible, meaning it never overestimates the real cost.

> *8-puzzle example: the number of misplaced tiles and the Manhattan distance are both admissible heuristics; Manhattan distance is more informed, so A* expands fewer nodes.*

Adversarial search

Minimax picks the move that maximizes the player's worst-case outcome in two-player games. Alpha-beta pruning skips branches that cannot influence the decision, giving the same result with far fewer evaluations.